

3. You are working with the BBC and want to develop an English-Yoruba translation model for BBC Monitoring. Yoruba is from the Edekiri family of languages and spoken widely in Nigeria and Benin. It is a very distant language from English, which is in the Germanic language family. What makes the task even harder is that there are very few freely available resources to support training an English-Yoruba translation model. You have managed to create a small test set of translated sentences from websites which you know contain English-Yoruba translations. But you would like to look for more English-Yoruba parallel sentences, and you have access to a large crawl of the internet.
- (a) What are the major steps to create a parallel English-Yoruba corpus from this crawl? [2 marks]
 - (b) Why is it especially difficult to find parallel data for a low-resource language pair from internet crawls? [1 mark]
 - (c) You have now found 20,000 sentences of parallel English-Yoruba data from the crawl, and you train a translation model on this data. Unfortunately looking at the output of your translation model, the quality is really poor. You go back to your crawl of the internet and extract all the monolingual Yoruba data, about 100,000 sentences. What could you do to improve your English-Yoruba translation model performance with this monolingual data? [4 marks]
 - (d) You decide to see if transfer learning from a high quality, high resource language pair would improve performance. There are no high resource related language pairs, so you select a German-English translation model to initialise your English-Yoruba system. How would you train a BPE subword level vocabulary for a system that covers English, German and Yoruba? What problem could occur with this multilingual BPE vocabulary? [2 marks]
 - (e) What are two advantages of BPE over modelling the vocabulary at the word level, with backoff to UNK? [2 marks]
 - (f) What are two ways of automatically evaluating the quality of your English-Yoruba system? Please briefly explain the each metric. [2 marks]
 - (g) Name one disadvantage of each of these automatic evaluation methodologies? [2 marks]
 - (h) Before deploying your system in the newsroom, what kind of human evaluation would be most valuable and why? [2 marks]